
TROPIC01

ODR_TR01_PSR_001_PUF_TropicSquare

TROPIC01 Physical Unclonable Function (PUF)

Version: 1.0

Abstract

This security report presents the results of a gray-box security and reliability evaluation of the physical unclonable function (PUF) integrated in the TROPIC01 secure element, performed in-house by Tropic Square. The PUF IP met the standard PUF quality metrics at every measured condition, with no degradation across the operating range.

June 3, 2026





1 Summary of the Security Report

1.1 Report Details

- **Evaluation Laboratory:** Tropic Square
- **Evaluation Laboratory Contact:** Tropic Square Security Team, security@tropicsquare.com
- **Evaluation Period:** Measurements captured 06/2025 - 09/2025
- **Type of Evaluation:** In-house evaluation by Tropic Square using Open Samples (custom samples prepared by Tropic Square).

1.2 Executive Summary

This report details a security and reliability evaluation of the TROPIC01 Physical Unclonable Function (PUF) performed by Tropic Square. To assess the quality and stability of the PUF responses, a gray-box evaluation was conducted on 300 chips across wafer corners. PUF measurements were collected from each device across a set of environmental conditions with temperatures ranging from -40°C to 125°C and PUF supply voltages between 1.212 V and 1.485 V.

The evaluation shows that the TROPIC01 PUF behaves according to expectations, performing in line with ideal theoretical models. Standard PUF metrics, specifically Uniformity, Bit-aliasing, and Uniqueness, were evaluated and the PUF showed good performance in these metrics. Additionally, a subset of the NIST SP 800-22 statistical test suite for randomness testing was applied to the PUF responses, confirming the expected randomness properties of the PUF output.

In terms of stability, the PUF maintained very high reliability across the vast majority of the test matrix. The chips behaved correctly under all tested conditions, showing consistent PUF response generation. However, two outlier chips exhibited a considerable number of bit flips that exceeded the capabilities of the error correction code used.

Apart from the two outlier chips, the data confirms that the overall PUF is robust and performs as expected within the defined operating conditions. The PUF is therefore suitable for the purposes used in TROPIC01.



1.3 Tropic Square Recommendations

Tropic Square regularly publishes FW updates to their products and resolves product weaknesses responsibly, as described in the Coordinated Vulnerability Disclosure Policy (CVD) (<http://tropicsquare.com/security/cvd>).

Version History

Version	Date	Description
1.0	3.6.2026	Initial Public Report Version



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2 The Evaluation Report

2.1 Evaluation Scope and Methodology

The aim of this report is to provide a comprehensive overview of the security and reliability evaluation process and results for the TROPIC01 physical unclonable function (PUF).

The PUF generates 256 raw stable bits after PUF evaluation. These bits are then hashed together with a user-provided 'challenge' value to create the final PUF response, which is used in different TROPIC01 subsystems [1].

To perform the PUF evaluation, only the raw stable 256 bits were measured from each chip, as the response creation is deterministic. These bits were read out using a debug interface (Virgin state [1]), which is disabled in production devices. This evaluation is therefore done in a gray-box context, where the raw PUF bits are accessible (unlike in the production devices), but no additional information about the PUF design or implementation is used. This enables Tropic Square to share this report publicly.

The goal of this evaluation is to assess the quality of the PUF response and the performance of the PUF across the operating conditions specified by the IP vendor.

The IP vendor claims the following operating conditions [2]:

Parameter	MIN	TYP	MAX
T junction	-25°C	25 °C	105°C
Supply voltage	1.212 V	1.35 V	1.485 V

2.2 Test conditions

For each chip, 1000x256 raw stable bits were collected for each of the listed test conditions. In total, 300 chips across wafer corners were used for the evaluation.

The measurements were carried out under the following environmental conditions:

- **Temperatures [°C]:** [-40, -25, 25, 105, 125]
- **PUF core supply voltage:** 'low' ~ 1.212 V, 'default' ~ 1.35 V (applied in production devices) and 'high' ~ 1.485 V



Note

Going forward, the 256 raw stable bits will be referred to as the 'PUF response' to maintain conventional nomenclature.

Note

The dataset was captured also outside the operating conditions of the PUF IP to test its performance for potential randomness degradation outside the specified operating conditions (-40°C and 125°C).

Note that these conditions are detected by the TROPIC01 temperature sensor on production chips.

The PUF output is considered under different standard PUF evaluation criteria:

1. Uniformity
2. Bit-aliasing
3. Uniqueness
4. Reliability
5. Randomness

To clarify the notation used in the subsequent section, we provide a definition table:

Table 1: Notation used in PUF evaluation parameter definitions.

Notation	Definition
k	Total number of devices ($k = 300$)
n	Number of bits in a response (256)
m	Number of response samples collected from one chip (1000)
$R_{i,j}(n)$	The j -th response (containing n bits) from chip i
$R'_i(n)$	The reference response of chip i , consisting of n bits
$r'_{i,j}$	The j -th bit of the reference response for chip i
$\text{HW}(x)$	Hamming weight of x
$\text{HD}(x, y)$	Hamming distance between x and y
$B(n, p)$	Binomial distribution with parameters n and p

The reference response $R'_i(n)$ is determined by a bitwise majority vote among m samples obtained under standard operating conditions. The standard operating conditions are:



1. temperature 25 °C
2. 'default' ~ 1.35 V supply voltage

For all performed statistical tests, a significance level of 1 % was chosen.



2.3 Uniformity

The uniformity U_i for a single response from chip i is calculated as the percentage Hamming weight of the response bits:

$$U_i := \frac{1}{n} \sum_{j=1}^n r'_{i,j} = \frac{\text{HW}(R'_i(n))}{n} \cdot 100 \% \quad (1)$$

The ideal value of uniformity is 50 %.

A statistical test is constructed to provide a threshold for the uniformity values. The test statistic used is the number of '1' bits in the response: $T = \text{HW}(R'_i(n))$. Under the null hypothesis that the response bits are independent and unbiased, $T \sim B(256, 0.5)$ is binomially distributed, where $p = 0.5$, is the probability of each bit being a '1' and $n = 256$ is the number of PUF bits. This yields a min and max threshold for the uniformity values on the 1 % significance level.

The visual distribution of uniformity values (Figure 1) aligns with the expected bounds. To rigorously verify this, a Pearson's chi-squared goodness of fit test was performed to confirm if the test statistic values T follow the expected binomial $B(256, 0.5)$ distribution (Figure 2). The test yielded a p-value = 0.064, exceeding the 1 % significance level.

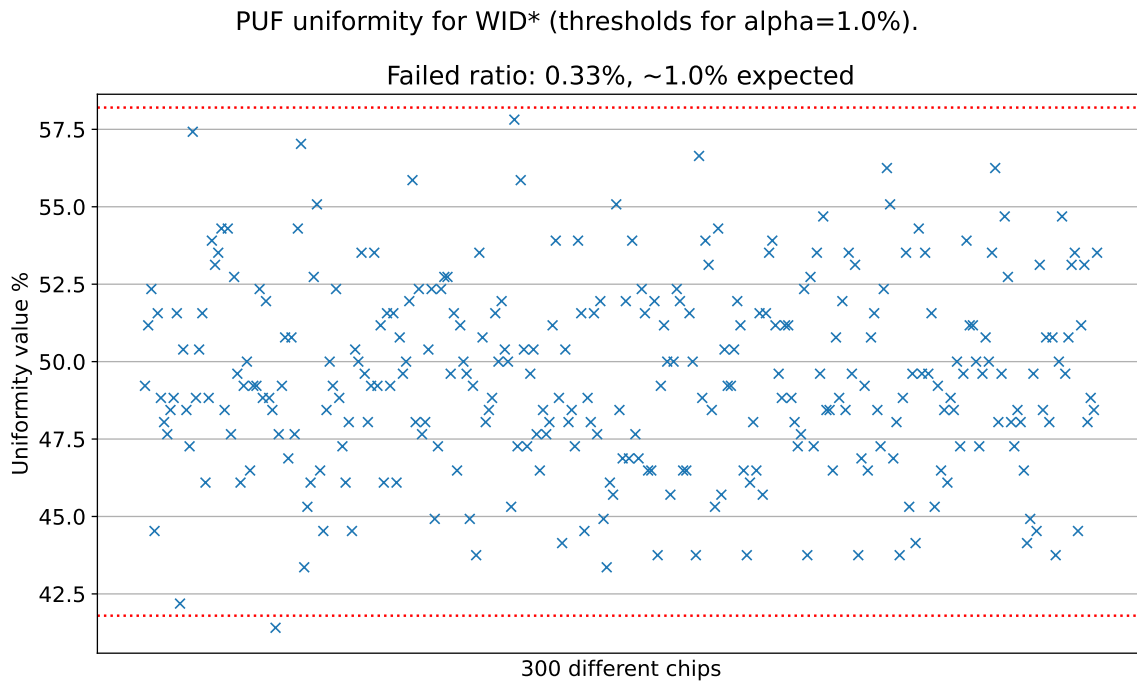


Figure 1: PUF uniformity values with 1 % thresholds.

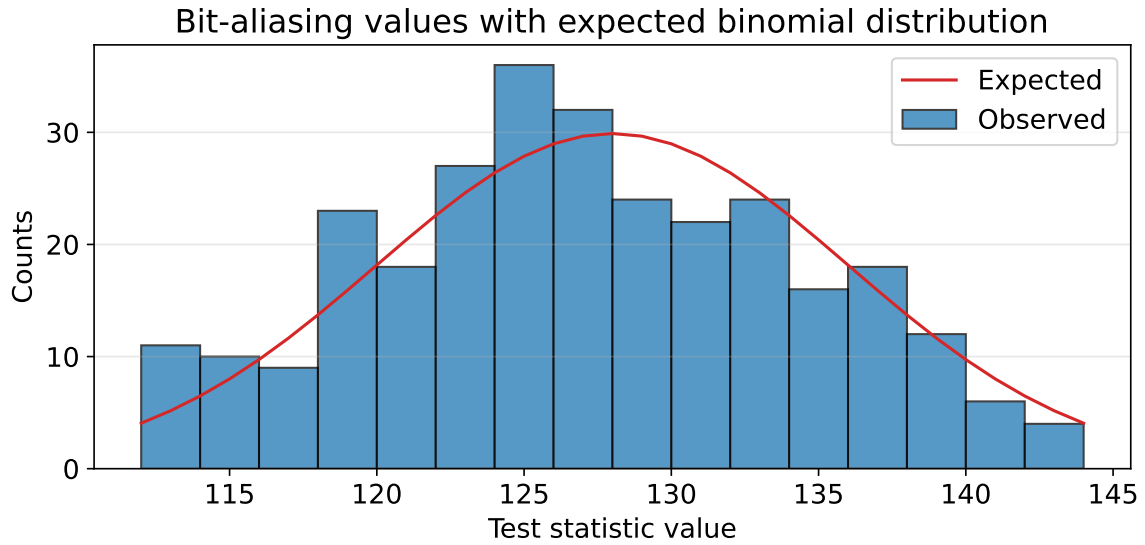


Figure 2: PUF uniformity test statistic values histogram with the expected $T \sim B(256, 0.5)$ distribution.

2.4 Bit-aliasing

Bit-aliasing is calculated by computing the percentage Hamming weight for the j -th bit:

$$\text{Bit-aliasing}_j := \frac{1}{k} \sum_{i=1}^k r'_{ij} \cdot 100\% \quad (2)$$

The ideal value of bit-aliasing is 50 %.

This metric is evaluated using the same statistical framework as uniformity, but rather than analyzing the response axis, bit-aliasing is calculated across the device axis. It is binomially distributed $T \sim B(n, 0.5)$, where n is the number of devices ($n = 300$).

As illustrated in Figure 3, the bit-aliasing metrics conform to the statistical thresholds. The subsequent chi-squared test for the test statistic values (Figure 4) yielded a p-value = 0.513 against the expected $T \sim B(300, 0.5)$ distribution, which is well above the 1% significance level.

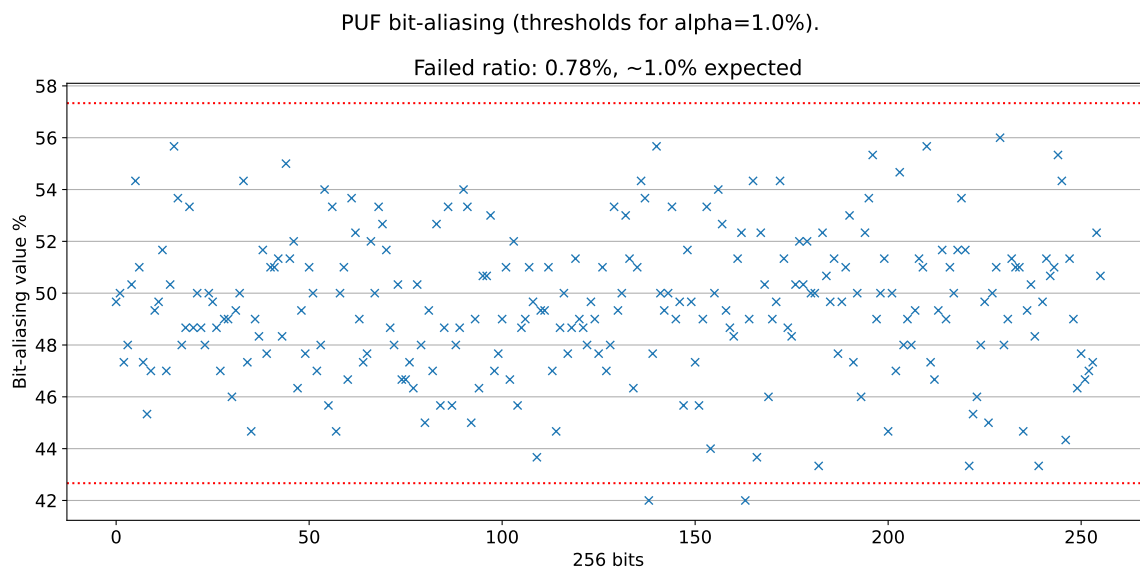


Figure 3: PUF bit-aliasing values with 1% thresholds.

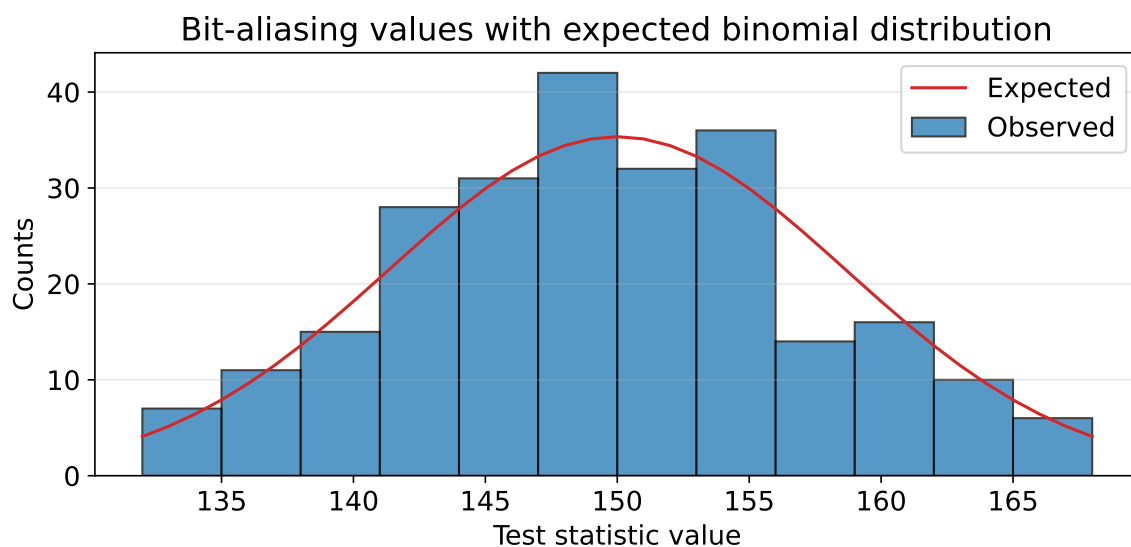


Figure 4: PUF bit-aliasing test statistic values histogram with the expected $T \sim B(300, 0.5)$ distribution.



2.5 Uniqueness

Uniqueness is calculated using the inter-Hamming distance:

$$\text{uniqueness} := \text{HD-inter} = \frac{2}{k(k-1)} \sum_{i=i}^{k-1} \sum_{j=i+1}^k \frac{\text{HD}(R'_i(n), R'_j(n))}{n} \cdot 100\% \quad (3)$$

The ideal value of uniqueness is 50 %.

The construction of a statistical test is similar to the one used for uniformity. The test statistic is the Hamming distance between two responses, $T = \text{HD}(R'_i(n), R'_j(n))$ (therefore $\binom{k}{2}$ test statistic values are obtained, one per pair of devices). Under the null hypothesis (bits of all responses across devices are independent and unbiased), $T \sim B(256, 0.5)$ is binomially distributed.

The plot (Figure 5) indicates that the uniqueness values are well within the defined thresholds. The chi-squared test on the uniqueness test statistic values against the expected $T \sim B(256, 0.5)$ distribution (Figure 6) supports this, yielding a p-value=0.066, which is larger than the 1 % significance level.

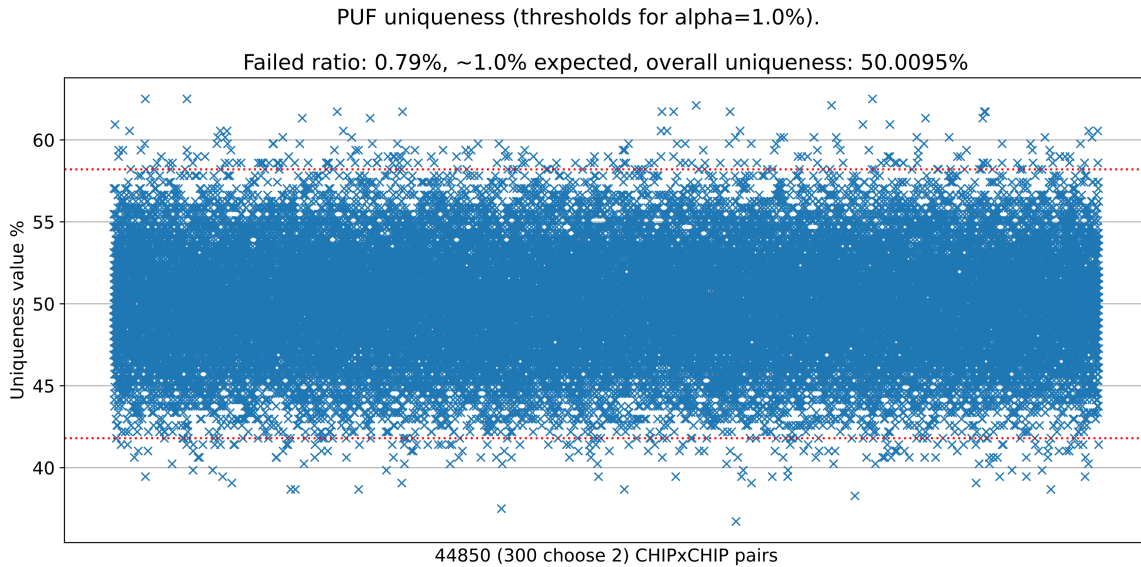


Figure 5: PUF uniqueness values with 1% thresholds.

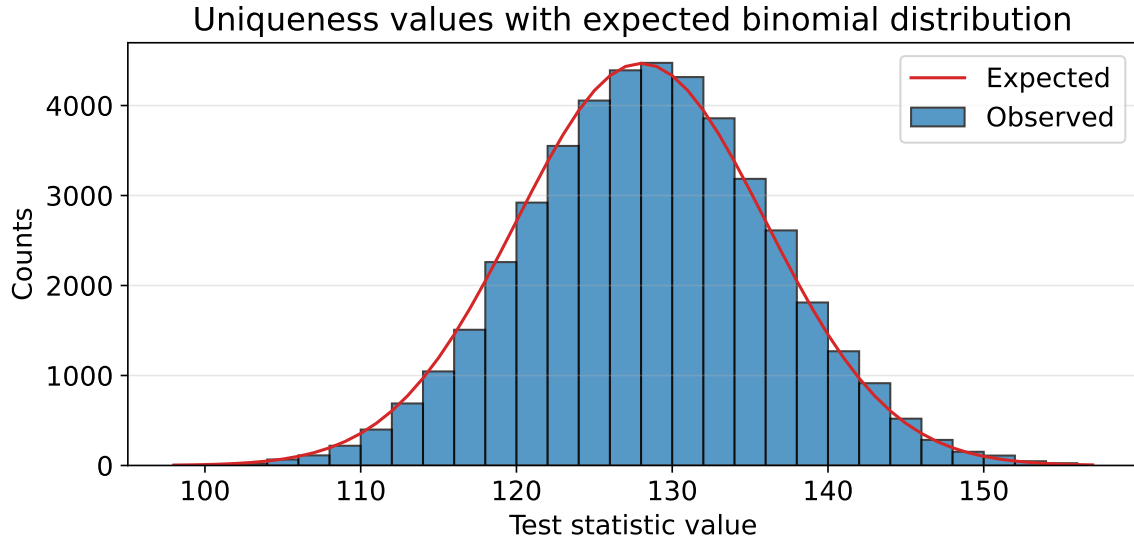


Figure 6: PUF uniqueness test statistic values histogram with the expected $T \sim B(256, 0.5)$ distribution.

2.6 Reliability

To calculate reliability, the intra-Hamming distance for a specific chip i is first needed:

$$\text{HD-intra}_i = \frac{1}{m} \sum_{j=1}^m \frac{\text{HD}(R_{i,j}(n), R'_i(n))}{n} \cdot 100\% \quad (4)$$

The reference response $R'_i(n)$ is recorded under standard operating conditions – room temperature and nominal voltage. Subsequent 1000 responses $R_{i,j}(n)$ are collected under varied conditions, and the intra-Hamming distance calculates the percentage of bits that differ from the reference.

Reliability is then calculated as:

$$\text{Reliability} = 100\% - \text{HD-intra} \quad (5)$$

We also evaluate a fail ratio. Fail ratio is calculated as the percentage of responses that differ from the reference response after error correction. This metric is useful as the PUF is required to always produce the same output in TROPIC01.



Note

The 256 PUF raw stable bits are protected by error correction code that can correct up to 2 bit errors. Fail is therefore considered if more than 2 bit flips happen - the case where the PUF output would in reality differ. This is why reliability and fail ratio are 2 separate metrics, as some loss in reliability is correctible.

We provide a heatmap of average reliability and average fail ratio (averaged values across all chips) for each environmental condition:

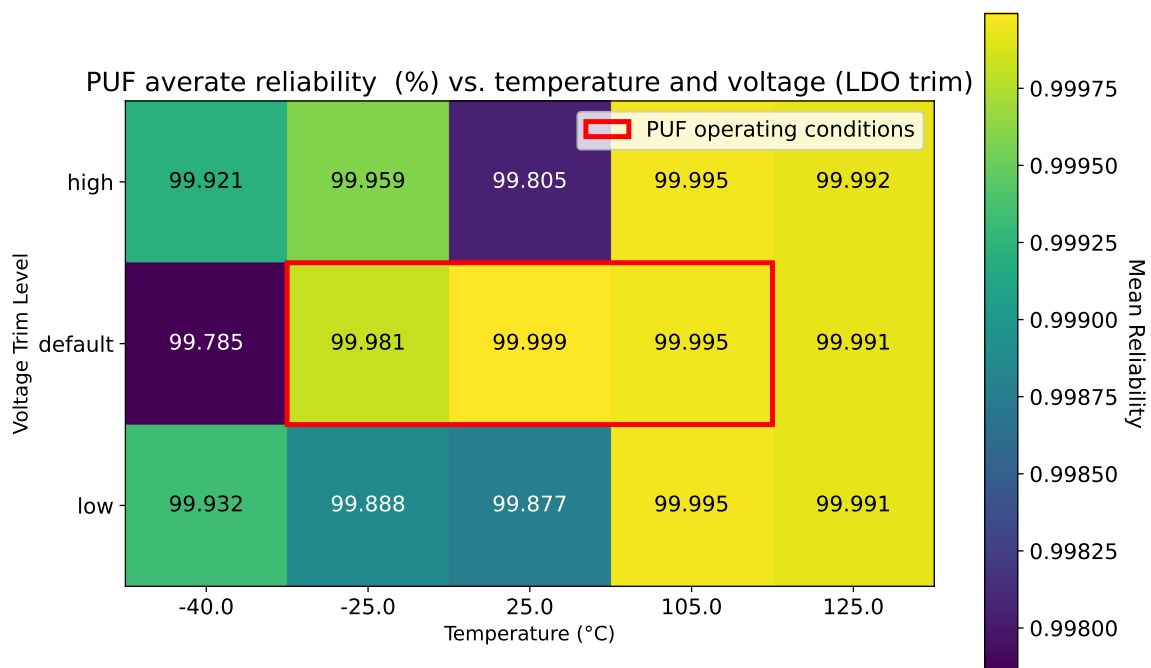


Figure 7: PUF reliability heatmap.

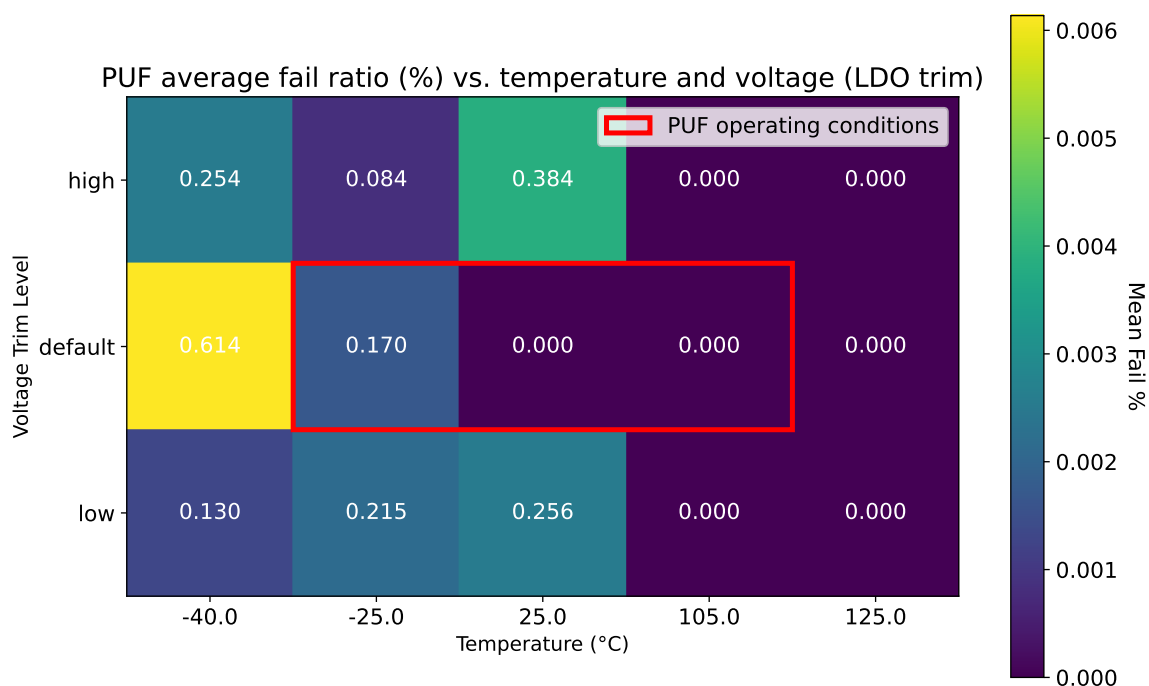


Figure 8: PUF fail ratio heatmap.

The reliability values in the PUF operating conditions are very high and do not decrease significantly even outside the described operating conditions.

However, we observe a non-zero fail ratio also in the PUF operating conditions.

Here is a table with the exact chips that have non-zero fail ratio in the operating conditions:

Table 2: Chips that have non-zero fail ratio in operating conditions

CHIP_ID	reliability %	fail ratio	temperature °C	trim
WID8_CID77	99.9695	0.001	-25	default
WID3_CID310	99.9461	0.001	-25	default
WID3_CID313	96.3715	0.074	-25	default
WID3_CID253	98.4742	0.434	-25	default

We see that the fails are only in the corner -25 °C temperature and the first two chips only failed 1 in a 1000 times.



However, chips WID3_CID313 and WID3_CID253 fail a considerable number of times and are considered outliers.

2.7 Randomness

Randomness will be tested using a subset of statistical tests from the NIST SP 800-22 statistical test suite (STS) [4]. The implementation used is a 'fast-sts' implementation from [5]. It also includes a Kolmogorov–Smirnov test for uniformity ($P\text{-value}(KS)$) as an addition to the chi-squared test in the standard results file.

Only a subset of the tests has to be used, as some tests require a huge number of bits as an input (1 million), but the PUF produces only 256 bits per chip.

The following tests will be used:

- Frequency (Monobit) Test
- Frequency Test within a Block
- Runs Test
- Test for the Longest Run of Ones in a Block
- Serial test
- Approximate entropy test
- Cumulative sums (Cusum) test

To see the exact parameters of the test, see [3].



In NIST STS terminology, a reference response for each device will be a sequence and the number of devices will be the number of streams (Each test will be run on each reference response separately).

The randomness results are interpreted using two distinct evaluation modes:

1. **Multiple Sequence Evaluation:** The responses from all devices are grouped as n independent streams and the testing is performed on the whole dataset. This allows the test suite to calculate results for the proportion test and assess the uniformity of p-values across all devices.
2. **Single Sequence Evaluation:** The statistical tests are interpreted for each device individually. The statistical test suite is run only on one (256 bit) sequence from each device, resulting in a PASS/FAIL result only for each test. This allows us to assess the randomness of the response on a per-device basis, but does not allow to calculate the proportion test or assess the uniformity of p-values.



The results for the **Multiple Sequence Evaluation:** are as follows:

RESULTS FOR THE UNIFORMITY OF P-VALUES AND THE PROPORTION OF PASSING SEQUENCES

generator is </tmp/tmpwyp1ht_a/data_file.txt>

C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	P-VALUE	P-value(KS)	PROPORTION	STATISTICAL TEST
29	24	42	30	20	26	34	29	25	41	0.082177	0.082386	0.9967	Frequency
36	18	30	38	24	29	33	30	34	28	0.324180	0.702758	0.9967	BlockFrequency
30	29	26	26	24	25	38	15	44	43	0.003046	0.007484	1.0000	CumulativeSums
30	19	32	39	32	21	25	22	35	45	0.012650	0.022607	0.9967	CumulativeSums
34	34	32	35	20	23	21	39	32	30	0.184884	0.298128	0.9900	Runs
19	37	32	33	27	39	24	27	34	28	0.257872	0.427105	0.9967	LongestRun
37	35	42	20	35	18	28	30	24	31	0.040108	0.037188	0.9933	ApproximateEntropy
22	36	25	23	38	39	34	34	23	26	0.127148	0.170800	0.9833	Serial
26	27	34	29	29	36	32	30	27	30	0.961593	0.908369	0.9900	Serial

The minimum pass rate for each statistical test with the exception of the
random excursion (variant) test is approximately = 0.972766 for a
sample size = 300 binary sequences.

For further guidelines construct a probability table using the MAPLE program
provided in the addendum section of the documentation.

We see that all the statistical tests passed in the proportion as well as in the uniformity
criteria.

Results from the **Single Sequence Evaluation:** can be seen in the following tables:

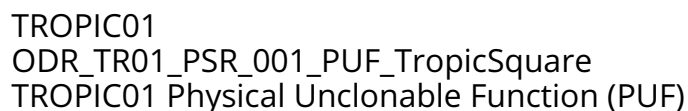


PUF randomness test results WID3					PUF randomness test results WID3				
WID3_CID224	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID226	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID228	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID229	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID230	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID232	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID237	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID238	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID242	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID247	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID248	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID253	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID255	-1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID256	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID259	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID260	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID261	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID263	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID265	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID268	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID269	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID271	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID274	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID275	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID277	-1.0	1.0	1.0	1.0	1.0	0.0	1.0	1.0	1.0
WID3_CID278	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID281	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID287	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID289	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID291	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID292	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID294	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID295	-0.0	1.0	1.0	1.0	1.0	0.0	1.0	1.0	1.0
WID3_CID296	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID298	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID299	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID300	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID301	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID304	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID305	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID308	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID312	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID316	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID317	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID318	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID322	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID323	-1.0	1.0	1.0	1.0	1.0	0.0	1.0	1.0	1.0
WID3_CID349	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID351	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
WID3_CID366	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
ApproximateEntropy									
BlockFrequency									
CumulativeSums									
Frequency									
LongestRun									
Runs									
Serial									

(a) First half of chips.

(b) Second half of chips.

Figure 9: Randomness tests results for WID3 chips. Overall failed ratio: 0.71%, 1.0% expected.



(a) First half of chips.

(b) Second half of chips.

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(a) First half of chips.

(b) Second half of chips.

Figure 11: Randomness tests results for WID11 chips. Overall failed ratio: 0.43%, 1.0% expected.

We see that the majority of the tests are passing with an expected number of false pos-



itives on the 1 % significance level.

2.8 Conclusions

All presented PUF evaluation parameters are in line with expectation, which was confirmed by statistical tests.

The presented data was calculated based on the 300 chips across all available corner splits.

However, the reliability metric had 2 outliers, namely the WID3_CID313 and WID3_CID253 chips. They showed high fail ratio at the -25 °C temperature corner. In other working conditions, all 300 chips behaved correctly.

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